Highlights

Applicability Boundaries and Fairness Challenges of BBR Congestion Control: A Three-Dimensional Framework, Version Evolution, and Multi-Scenario Evaluation

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- BBR evidence is organized by path conditions and implementation.
- An analytical assessment separates utilization from bandwidth allocation.
- Controlled experiments span 340 runs in four implementation environments.
- Comparable utilization coexists with sharply different bandwidth shares.
- Service requirements determine joint goodput and fairness assessments.

Applicability Boundaries and Fairness Challenges of BBR Congestion Control: A Three-Dimensional Framework, Version Evolution, and Multi-Scenario Evaluation

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ABSTRACT

BBR can sustain high aggregate throughput while competing flows receive unequal bandwidth shares. Interpreting this tradeoff requires the network conditions, controller implementation and performance objective to be considered together. This survey connects BBRv1–v3 evidence across six network scenarios through buffer depth, bandwidth-delay product and loss source, with implementation, queue management and flow count as assessment inputs. An analytical procedure separates utilization from allocation and expresses fairness requirements as explicit flow-share constraints. We complement the synthesis with 340 controlled two-flow runs covering four implementation environments, RTT asymmetry, BBR/CUBIC coexistence, additional capacity and delay settings, and artificial loss. In the twelve BBR/CUBIC conditions, total goodput is 95.58–95.59 Mbps while the BBR share spans 0.143–0.327. Additional equal-RTT comparisons show how changing the goodput or fairness requirement changes the assessment of BBRv3 and CUBIC. Together, the synthesis and measurements provide a reproducible, metric-specific basis for evaluating BBR and interpreting its deployment tradeoffs. Learning-assisted approaches are assessed through their control objectives, reported outcomes and deployment evidence.

1. Introduction

1.1. Background and Motivation

Building on Jacobson's congestion avoidance algorithm [1], loss-driven congestion control—represented by CUBIC [2] and Reno [3]—has dominated the Internet transport layer for decades. Loss-driven flows can build queues while probing available capacity, placing their operating point beyond the just-full-pipe region discussed in Kleinrock's optimal power analysis [4]: queueing delay can increase, depending on buffering, load and queue management. Large bandwidth-delay products (BDP) also make receiver-window and loss-recovery constraints important. The Reno-like throughput approximation of Mathis et al. ($\mathrm{BW} \propto \mathrm{MSS}/(\mathrm{RTT}\sqrt{p})$) [5] illustrates the sensitivity to RTT and loss under its steady-state assumptions. CUBIC has a different window evolution, so this approximation cannot serve as its universal throughput bound. Google proposed BBR (Bottleneck Bandwidth and Round-trip propagation time) in 2016 [6], replacing loss-signal-driven reactive control with direct measurement of the bottleneck bandwidth BtlBw and propagation delay RTprop, and demonstrating significant throughput advantages over CUBIC in large-scale deployments on Google's B4 WAN. BBR subsequently released BBRv2 in 2019 (introducing ECN support and in-flight upper-bound constraints) [7] and BBRv3 in 2023 (further tightening Startup exit conditions) [8]. Unlike BBRv1, which has entered the Linux mainline, BBRv2/BBRv3 continue to evolve primarily through drafts, implementation prototypes, and independent evaluations. Meanwhile, QUIC/HTTP 3, LEO satellite measurements, and CDN scenario studies have expanded BBR's influence beyond the traditional TCP stack [9–13].

Published evaluations show configuration-dependent throughput and fairness. In WLAN studies, pacing and sender queue constraints interact with frame aggregation; comparisons require the tested chipset, direction and queue configuration. RTT fairness likewise depends on probing, path estimates, queue

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policy and competing flows. The BDP identity describes an in-flight operating point and does not determine bandwidth shares. ECN availability changes the feedback signals available to BBRv2/v3 but does not erase their loss-driven updates or version-specific logic. Multi-RTT simulation results illustrate conditional differences among versions rather than a universal ranking [14–18]. AI-assisted improvements, though demonstrating promise in controlled testbeds or simulation environments, require evidence on interpretability, transfer outside their training conditions and real-time inference overhead.

Comparisons across the literature are difficult because experiments differ in capacity, delay, queue policy, implementation and workload. A result reported for one configuration may not answer an operator's question about another. BBR's use in Linux, QUIC and satellite-link research makes these conditions relevant to deployment assessment [6, 7, 19–21]. We organize the evidence around explicit path and implementation inputs, then examine which metric-specific conclusions those inputs and the available measurements support.

1.2. Related Surveys and Positioning

Existing surveys and evaluations examine BBR's performance, deployment conditions and fairness. Their findings need to be interpreted within the versions and network configurations they study. Early independent evaluations focused primarily on BBRv1's performance in controlled wired scenarios: Hock et al. [22] revealed RTT bias, Cao et al. [23] identified the random loss "cliff effect," and Jaeger et al. [24] and Scholz et al. [25] provided reproducible experimental baselines. Recent BBRv3 research has pushed the problem toward fairness and deployment boundaries: Piotrowska [18] reported that v3's intra-protocol fairness regresses relative to v2 in large-buffer scenarios, Gomez et al. [26] provided real-world broadband baselines, and Zeynali et al. [21] and Bless et al. [27] further revealed extreme unfairness and self-induced queuing on public Internet paths and high-speed links.

This survey connects version-specific mechanisms with the conditions under which throughput and fairness are measured. Earlier measurements and surveys provide the source evidence; our organizing contribution is to report buffer depth, BDP and loss source together with implementation, queue policy and flow count. The added controlled experiments compare these conditions explicitly. Table 1 compares the scope and organizing approach of the reviewed studies. The framework connects performance assessments to explicit path, implementation and workload conditions.

Related BDP-aware transport designs also include BDP-Veno and OVeno [29, 30]. We include them as related approaches rather than BBR variants. Their shared use of BDP does not establish an identical bandwidth estimator, probing policy or loss response, and we do not infer a performance ranking without matched experimental conditions.

1.3. Literature Search and Evidence Verification

This article combines a critical narrative synthesis with a documented supplemental search. We separate the original reference collection from search-derived eligibility decisions. The original 165 citations were

Table 1
Comparison with existing BBR-related surveys and systematic evaluations.

Work	Type	Ver.	Scenarios	Fairness	Framework	AI	Sat.	Wireless	Methodol.
Hock et al. [22] (2017)	Meas.	v1	1	Partial	None	No	No	No	No
Cao et al. [23] (2019)	Meas.	v1	2	Partial	None	No	No	No	No
Jaeger et al. [24] (2019)	Meas.	v1	3	✓	None	No	No	No	No
Al-Saadi et al. [28] (2019)	Survey	v1	2	Partial	None	No	No	No	No
Piotrowska [18] (2024)	Simul.	v1–v3	1	✓	None	No	No	No	No
Gomez et al. [26] (2024)	Meas.	v3	1	✓	None	No	No	No	No
Abrol et al. [20] (2026)	Survey	v1–v3	4	✓	None	No	No	5G	No
Zeynali et al. [21] (2024)	Meas.	v3	1	✓	None	No	No	No	No
Bless et al. [27] (2025)	Meas.	v3	2	Partial	None	No	No	No	No
This paper	**Survey**	**v1–v3**	**6**	✓	**3-D**	✓	✓	✓	✓

"Methodol." indicates whether an explicit literature search protocol is described; "5G" in the Wireless column indicates coverage of 5G scenarios only.

a legacy reference collection; their historical retrieval and screening chain was not retained. We do not reconstruct that chain from citation presence. Instead, the revision provides a retrospective, citation-level identity and role register for every original entry, with title, year, DOI or URL, source-record matches and the limits of available verification.

Table 2 distinguishes directly relevant BBR material from related methods and supporting context. Roles are assigned from titles, metadata and manuscript use, independently of whether a reference was found by the supplemental queries. They are not full-text inclusion decisions or independent study counts. Exact DOI matching locates 41 entries in the supplemental exports; seven further entries have exact normalized-title candidate matches. The other 117 are not located by either rule in these capped exports. A title candidate does not confirm identity, and absence from a capped search does not establish irrelevance. Each legacy reference retains its known acquisition route with unknown historical routes identified explicitly.

Existing citation is not an eligibility criterion for the supplemental search. The supplemental protocol was frozen on 28 September 2026. Eight query groups in OpenAlex and Crossref cover BBR versions, fairness, buffering, loss and ECN, related BDP-aware transport, wireless and satellite paths, and learning-based control. The exports contain 1,018 OpenAlex and 1,600 Crossref records (2,618 in total); retrieval was capped at 200 records per query, and 12 of 16 responses exceeded that cap. DOI normalization and cross-source matching yield 1,916 report groups. The search is a bounded supplement and does not establish exhaustive coverage. Figure 1 reports the retained retrieval and screening stages. The accompanying search supplement preserves database-specific counts, queries, deduplication records and exclusion reasons.

Automated title/abstract triage was used to prioritize records, not to establish inclusion. Subsequent relevance reassessment and full-text checks yielded 171 title/abstract exclusions and 15 completed full-text assessments (8 retained and 7 excluded); 1,730 report groups remain unassessed at full text. These report-level counts do not constitute a completed review corpus. We retain this limitation explicitly and make no claim of exhaustive coverage or predictive validation based on citation counts.

Eligibility concerns BBR mechanisms, performance and fairness, related transport designs, network conditions, and supporting standards and methods. Source assessment distinguishes simulation, emulation and deployment evidence and records implementation, parameters, observation windows and result locations. Quantitative comparisons are interpreted within their reported conditions; unavailable full texts and incomplete appraisals limit the synthesis. Source assessments retain these conditions alongside each quantitative comparison.

The cited references are grouped thematically into five areas: BBR model boundary foundations [31–59], data center and programmable scheduling boundaries [60–96], BBR version and independent evaluation [97–107], QUIC and deployment scenarios [108–127], and learning-based congestion control and experimental platforms [106, 128–137].

To further constrain quantitative claims, Table 3 lists the evidence sources and applicable conditions for the paper's core numerical findings. Beyond data explicitly reported in the original papers, we do not extrapolate values from public talks, drafts, or secondary sources into general conclusions. Heterogeneity across studies—arising from differences in testbed type, statistical metrics, AQM configuration, and BBRv1 kernel version—is particularly notable for AQM configuration: Shrestha et al. [138] showed that FQ-CoDel can raise the inter-flow Jain fairness index between BBRv3 and CUBIC from approximately 0.6 to approximately 0.9. We annotate AQM configuration information in relevant scenario analyses; for papers that do not report configuration, caveats on the generalizability of their conclusions apply.

1.4. Contributions and Paper Organization

The survey addresses three connected questions: which path and implementation conditions are needed to interpret BBR results; which mechanism claims can explain measured allocation without mistaking a rate–delay identity for a share law; and how evidence can support a metric-specific performance assessment.

Our first contribution is a conditional organization of BBRv1–v3 evidence across six scenario groups. It separates direct BBR evaluations, mechanism materials and supporting transport context, and supplies a reference-level traceability register and the retained supplemental-search records. This differs from a coverage-only comparison by keeping the conditions needed to interpret throughput, delay, fairness and retransmissions. Bringing these scenarios into a common assessment reveals which inputs and metrics are needed to interpret results across configurations.

Table 2
Retrospective roles of the 165 original references. The accompanying register gives the assignment basis and source matches for each entry. These citation-level roles are not a systematic-review inclusion count.

Role	References
Direct BBR evaluation or analysis	34
BBR mechanism or version material	5
Related surveys	2
Learning/control comparators	15
Supporting transport/network context	109
Total legacy references	165

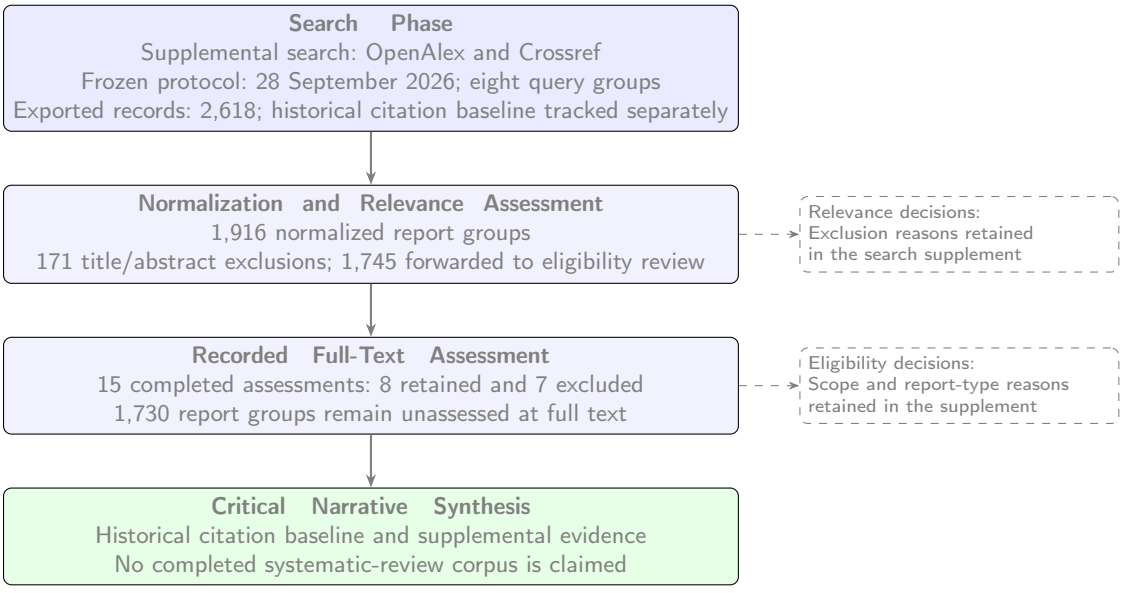


Figure 1: Recorded supplemental-search stages accompanying the narrative survey. Counts are report-level; the supplemental search is capped and full-text coverage is incomplete.

Our second contribution is an analytical assessment procedure (Section 3.5). It demonstrates the non-identifiability of bandwidth shares from the in-flight identity, expresses two-flow fairness requirements as share constraints, and gives explicit requirement-dependent outputs including undetermined cases. The underlying BDP and Jain identities are established; the contribution is their use in a reproducible assessment with stated inputs, windows and service requirements.

Our third contribution is a 340-run controlled comparison with implementation identities, measurement definitions and reproducible data. Besides RTT asymmetry, BBR/CUBIC coexistence, additional equal-RTT cases and artificial loss, we quantify a retrospective transfer test from E1 calibration to disjoint E3 runs. Prediction errors against a constant baseline quantify the dependence of transfer estimates on queue conditioning. The learning-control discussion separates objectives, measured outcomes and deployment evidence.

Section 2 presents the controller foundations and version differences. Section 3 defines the framework and analytical assessment. Sections 4 and 5 synthesize scenario and fairness evidence; Section 6 examines learning-assisted control. Section 7 gives the experiments and transfer evaluation, and Section 8 states the resulting conclusions and limits.

Table 3
Source and condition mapping for scenario comparisons.

Reported comparison	Source	Applicable conditions
Backbone throughput improvements reported under source-specific path and host conditions	Cardwell et al. [6]	Google production WAN with open receive buffers; not generalized to all public paths
BBR shows throughput degradation or queuing anomalies under shallow-buffer/high-loss conditions	Cao et al. [23], Jaeger et al. [24]	Valid within controlled testbed parameter ranges; requires interpretation with buffer and loss source
BBRv3/CUBIC throughput depends on the evaluated configuration	Gomez et al. [26]	Real broadband testbed with reported AQM/path configuration; not representative of all broadband ISPs
BBRv3/CUBIC coexistence can exhibit strongly unequal shares	Abrol et al. [20], Zeynali et al. [21]	Extreme fairness cases in large-buffer or specific real paths; used to illustrate risk boundaries
Standard BBR may underperform CUBIC in WiFi frame-aggregation; BBRp partially mitigates loss	Grazia et al. [139]	802.11n/ac test conditions; not directly extrapolated to all WiFi 6/7 deployments
BBR outperforms CUBIC on Starlink-class LEO links; QUIC+BBR handover optimization	Deutschmann et al. [12], Kamel et al. [13]	Specific satellite, geography, terminal, and handover mode; independent replication still needed

2. BBR Algorithm Foundations and Version Evolution

2.1. Measurement Model and Sending Control

Understanding BBR's applicability boundaries requires first understanding the network model on which its design rests—BBR's advantages and limitations both stem from the same set of measurement assumptions.

BBR models the path using estimates of bottleneck bandwidth (BtlBw) and propagation delay (RTprop). Their product provides a reference in-flight volume for the "just filling the pipe" operating target [4, 6]. In the BBRv1 design, delivery-rate samples feed a maximum filter over ten packet-timed rounds, and RTT samples feed a minimum filter with a ten-second window. These estimates can be affected by application or receiver limits, ACK timing, queueing and path changes. Filter horizons and update rules must be checked against the particular version and implementation rather than transferred unchanged to BBRv2 or BBRv3 [6, 7, 48]. Figure 2 illustrates this operating target and the underutilization/bufferbloat trade-off around the BDP point.

A classical approximation for a Reno-like additive-increase, multiplicative-decrease flow is

$$\mathrm{BW}_{\mathrm{Reno}} \approx \frac{K \cdot \mathrm{MSS}}{\mathrm{RTT} \cdot \sqrt{p}}. \tag{1}$$

Here p denotes loss probability and K depends on modeling assumptions, including acknowledgment behavior [5]. The approximation describes a restricted steady-state regime and does not account for all timeout, receiver-window or application limitations. CUBIC has a different window evolution [2]. The Reno-like approximation motivates examining RTT and loss jointly within its stated steady-state regime.

A simplified BBRv1 model relates pacing and the target congestion window to these estimates:

$$\mathrm{pacing_rate}_{\mathrm{target}} = \mathrm{BtlBw} \times \mathrm{pacing_gain}$$

$$\mathrm{cwnd}_{\mathrm{target}} \approx \mathrm{BtlBw} \times \mathrm{RTprop} \times \mathrm{cwnd_gain}.$$

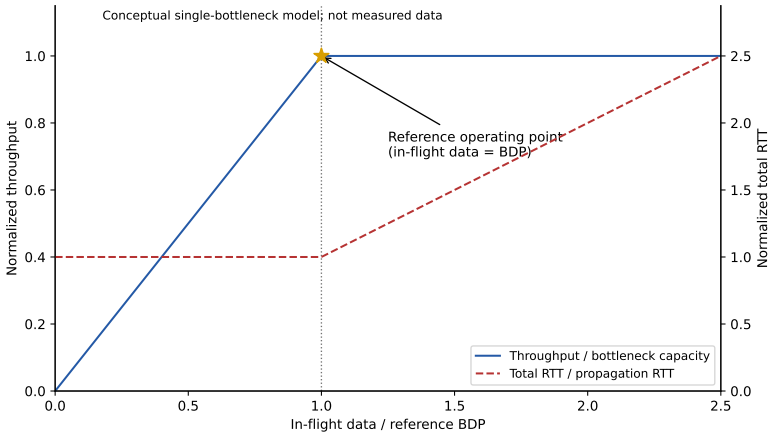


Figure 2: Conceptual operating point for a single bottleneck. Throughput saturates when in-flight data reaches the reference BDP. Propagation RTT is held constant; additional queue occupancy increases total RTT through queueing delay. Curves are illustrative, not measured data.

These are controller targets, not exact identities for the instantaneous Linux congestion window. Recovery, minimum-window rules, packetization, ACK aggregation and host limits can change the realized behavior. Pacing spreads transmissions in time [140], while the congestion window limits outstanding data [141]; queue overflow depends on their interaction with the available buffer.

The BBRv1 design has four principal modes (Figure 3). Startup seeks additional delivery rate; its full-bandwidth test checks whether estimated bandwidth has grown by at least 25% over three consecutive packet-timed rounds. Drain reduces the excess in-flight data accumulated during Startup. ProbeBW varies pacing gain to probe and drain, and ProbeRTT temporarily reduces the congestion window to obtain a less queue-inflated RTT sample [6]. These modes do not all recur periodically, and ProbeRTT does not guarantee a queue-free sample. Later versions revise the probing and in-flight constraints. ECN response must likewise be identified from the specific implementation and actual configuration; support for an ECN mechanism does not prove that it was enabled in a reported experiment [7, 14].

2.2. Core Differences Across Three Versions

BBR's evolution from BBRv1 in 2016 [6], through BBRv2 in 2019 [7], to the BBRv3 draft in 2023 [8], retains a delivery-rate and propagation-delay model while revising congestion responses and probing. The comparison below covers three mechanism groups: tightening Startup exit conditions, restructuring ProbeBW, and tuning ProbeRTT parameters. Figure 4 provides the visual summary, while Table 4 compares the key parameters.

Startup Exit Conditions BBRv1's full-bandwidth test uses insufficient estimated bandwidth growth across non-application-limited packet-timed rounds. Later versions add congestion-related exit conditions [7, 8]. In the inspected BBRv3 source, the loss-based check requires a Recovery round boundary, at least six loss-marking events in that round and an excessive in-flight loss indication. Six events are not simply six lost packets. The ECN-based check requires an eligible connection and two consecutive rounds meeting the configured CE-ratio threshold. These checks cannot be reduced to an unconditional packet-count trigger or a claim of improved fairness without measurements.

ProbeBW Structure BBRv1 uses an eight-phase gain sequence $(1.25, 0.75, 1.0\times6)$. Phase duration depends on elapsed minimum-RTT time, in-flight data and loss conditions; it is not exactly eight packet-timed rounds. BBRv2/v3 restructure this into four sub-phases—Refill/Up/Down/Cruise—introducing two dynamic bounds inflight_hi and inflight_lo [7]. The inspected BBRv3 source sets the Down gain with the integer expression $\lfloor 256\times91/100\rfloor/256 = 232/256$. This gives the source-specific drain gain [8]; coexistence is evaluated through measured per-flow rates.

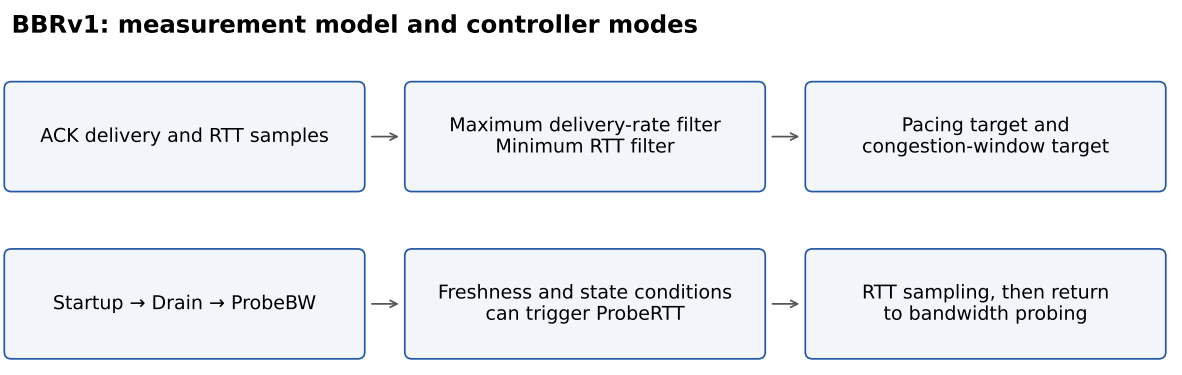


Figure 3: Simplified BBRv1 measurement and control architecture. Delivery-rate and RTT samples inform pacing and congestion-window targets. ProbeRTT entry depends on freshness and state conditions, rather than a fixed periodic timeout. Later-version ECN-bound mechanisms are not part of this BBRv1 diagram. The original figure was generated with GPT Image 2.0 and manually checked by the authors; this replacement was drawn with Python/matplotlib using AI-assisted code preparation.

ProbeRTT ProbeRTT entry depends on the implementation's RTT-sample freshness and state conditions; a filter window is not an unconditional periodic timer. The inspected BBRv3 source distinguishes a ten-second minimum-RTT filter from a five-second ProbeRTT sampling window, with a half-BDP target subject to the minimum congestion window. These parameters do not by themselves establish how much bandwidth a competing CUBIC flow obtains. The exact source revision and runtime conditions must accompany a version-specific comparison [7, 8].

QUIC implementations can also use BBR. QUIC runs over UDP and lets applications implement congestion control without changing the kernel TCP controller [9]. BBR's behavior specification in QUIC is now maintained by the IETF CCWG working group draft [19]. However, the QUIC+BBR combination also introduces new fairness issues, as multiple QUIC connections are treated by the OS as independent UDP flows whose aggregate behavior differs from native TCP flows [20].

Version mechanisms and conditions for comparison

Aspect	BBRv1	BBRv2	BBRv3
Path model	Delivery-rate and minimum-RTT estimates	Delivery-rate and minimum-RTT estimates	Delivery-rate and minimum-RTT estimates
Probing	Eight-phase ProbeBW sequence	Refill / Up / Down / Cruise	Revised four-phase probing
Congestion feedback	Model-based rate control; TCP recovery remains active	Loss-based bounds; ECN response when enabled/eligible	Loss-based bounds; ECN response when eligible
Parameters	Reference Linux v5.4 design	Defaults depend on alpha revision and runtime module settings	Defaults depend on source revision and eligibility conditions
Performance	Evaluate under matched conditions	No universal fairness ranking	No universal fairness ranking

Mechanism overview; not a measured performance ranking. Exact parameters are reported in the manuscript version-parameter table.

Figure 4: Version-specific mechanisms and comparison conditions. The overview does not imply monotonic improvement in throughput, delay or fairness; source-specific parameters are reported in Table 4. The original figure was generated with GPT Image 2.0 and manually checked by the authors; this replacement was drawn with Python/matplotlib using AI-assisted code preparation.

Table 4
Key parameter comparison of BBRv1/v2/v3 [6–8, 20].

Parameter	BBRv1	BBRv2	BBRv3
Startup exit	BW gain $<25\%$ (3 rounds)	BW-growth test plus conditional loss/ECN exits; 8 loss events in inspected alpha source	BW-growth test plus conditional loss/ECN exits; 6 loss events in inspected source
ProbeBW structure	8-phase sequence (1.25, 0.75, 1.0×6)	Refill/Up/Down/Cruise sub-phases	Four sub-phases; Down gain is revision-dependent
ProbeRTT target	4 MSS	$\max(4\,\mathrm{MSS}, \mathrm{BDP}/2)$	$\max(4\,\mathrm{MSS}, \mathrm{BDP}/2)$
ProbeRTT freshness window	10 s (v1 design)	5 s (inspected alpha source)	5 s (inspected source)
$\mathrm{cwnd_gain}$	2.0 (baseline)	State-dependent	State-dependent
$\mathrm{pacing_gain}$ (Down)	0.75	0.75	$232/256$ (inspected source)
ECN support	None	Implemented; disabled by default in inspected alpha source	Conditional on ECN eligibility
$\mathrm{inflight_hi}$ constraint	None	Yes	Yes
Primary goal	Baseline	Fairness & ECN	Reduce queue; relieve starvation

For BBRv2/v3, ECN marking provides an additional network-side signal (RFC 3168 [14]). In both inspected source snapshots, the loss counter counts marking events, not lost packets. These source defaults are distinguished from the identities of the experimental builds. ECN sender eligibility additionally checks negotiated capability and an RTT condition. Filter and freshness windows are not unconditional ProbeRTT entry intervals. When ECN marking is unavailable, ECN-derived feedback is absent, but loss-driven bound updates and the associated state-control logic remain active. The absence of ECN alone therefore does not imply reversion to BBRv1; parameter values and behavior remain implementation- and state-dependent.

3. Three-Dimensional Applicability Framework

BBR measurements differ across path conditions and implementations. This section organizes those conditions using buffer depth, BDP magnitude and loss source, with implementation, queue management and competing flows reported alongside them. The framework connects these inputs to separate throughput, delay, fairness and retransmission requirements. Measured flow rates provide the allocation evidence, while the analytical assessment and quantitative transfer comparison show how the inputs are interpreted.

3.1. Framework Design Motivation

BBR's strong dependence on network conditions indicates that a single-dimensional performance metric cannot effectively predict its behavior in unknown environments. Measurements by Medina et al. [49] show that transport protocol behavior varies greatly with network environment evolution; Barakat et al. [50] further reveals that TCP behavior is highly dependent on path characteristics in heterogeneous networks. We therefore propose a BBR applicability boundary analysis framework composed of three related physical dimensions, mapping BBR's behavioral characteristics under different network conditions to a unified coordinate space.

3.2. Three-Dimensional Definitions and Physical Meaning

The three dimensions reflect BBR's path model and the network conditions that affect its estimates and control responses. Their working numerical bins organize the evidence; they are not derived physical laws.

Buffer depth (measured in BDP units; this paper uses working bins of shallow $< 1.5\times$BDP, medium 1.5–10$\times$BDP, and deep $> 10\times$BDP) is an analysis dimension rather than a universal physical boundary. The theoretical basis comes from Kleinrock's optimal power theory [4]: BBR targets "just-full pipe" sending, periodically reducing cwnd during ProbeRTT to obtain queue-free RTprop estimates, and the relative cost of this active yielding depends on queue depth, RTT, competing traffic and implementation. Shallow buffers can expose probing-induced loss, while larger buffers can increase queueing delay. Neither buffer depth nor ProbeRTT alone fixes the resulting bandwidth shares [43–45]. Queue management and per-flow scheduling, including FQ-CoDel, change these interactions [142].

BDP magnitude (measured by the BW $\times$ RTT product; the working bins in this review use low < 1 MB and high > 10 MB) organizes capacity and delay conditions. Paths with the same BDP may have different RTTs, capacities, loss processes and controller behavior. The Reno-like approximation in Eq. (1) therefore does not establish a BDP-only throughput ceiling or a guaranteed advantage for BBR.

Loss source (congestion-induced / random / mixed) matters because packet loss alone does not identify its cause. BBRv1 primarily regulates sending through its delivery-rate and propagation-delay model, while BBRv2 and BBRv3 also use loss-related constraints. TCP recovery remains necessary in all versions. Random loss can reduce delivered goodput and increase retransmission overhead; its effect depends on implementation, queueing, link-layer recovery and workload. The WiFi experiments of Grazia et al. [139] further show why sender location, chipset and host queue settings must accompany a wireless comparison. Loss assessment combines controller responses with link-layer recovery, queue policy and workload.

Theoretical Basis for Thresholds The threshold values are working bins anchored to prior measurements and theoretical reasoning, rather than universal cutoffs. The buffer-depth bins summarize patterns reported by Cao et al. [23] and Jaeger et al. [24]; their effects depend on version, queue management, RTT, and flow count. BDP magnitude thresholds of 1 MB (low) and 10 MB (high) are also working bins, not universal boundaries. They organize the discussion and are not derived as performance transition points from Eq. (1). A numerical transition requires evidence for the specified version, workload and path.

3.3. Six-Scenario Positioning

Table 5 maps the six representative scenarios covered in this paper to the three-dimensional coordinate system.

3.4. Decision Rules and Known Limitations

The framework provides a condition-specific assessment procedure with explicit inputs and separate performance outputs. Its inputs are bottleneck capacity, the reference RTT used for BDP, absolute buffer

Table 5
Positioning of six network scenarios in the three-dimensional applicability boundary framework.

Scenario	Buffer depth	BDP	Loss source	Scope	Key constraint
High-BDP WAN	Medium	High	Congestion	Throughput	Recv. buffer $\geq$ BDP
Shallow-buf. wired	Shallow	Low	Congestion	Loss cost	Retransmission surge
Broadband multi-flow	Medium	Medium	Congestion	Coexistence	Fairness varies with queue
High-loss wireless	Medium	Low–Med	Random	Link/host	WiFi frame aggregation
High-dynamic routing	Medium	Medium	Mixed	Adaptation	Multi-path fairness unverified
LEO satellite	Variable	Varies	Mixed	Dynamics	RTT spikes, BW asymmetry

size, loss source, BBR version, queue management and flow count. For heterogeneous flows, the reference RTT and each flow's RTT are reported separately. Its outputs are four separate assessments: throughput, delay, bandwidth fairness, and retransmission cost. A missing input or insufficient evidence produces "undetermined" rather than a forced rating. The working buffer bins are shallow $< 1.5\times$BDP, medium 1.5–$10\times$BDP, and deep $> 10\times$BDP; these bins organize the reviewed evidence and are not universal physical thresholds.

Based on these conditional rules, Table 6 summarizes scenario-level expectations and known counter-examples. The entries are evidence summaries, not guarantees for an arbitrary deployment.

Table 6
Three-dimensional framework decision rules and known counter-examples.

Typical condition	Buf.	BDP	Loss source	Conditional expectation
High-BDP backbone (recv. buf. $\geq$BDP)	Med	>10 MB	Cong.	Throughput advantage is plausible when receive buffer $\geq$BDP; delay, fairness, and retransmission require separate checks
Wired multi-flow (1.5–$10\times$BDP buf.)	Med	Med	Cong.	Throughput may improve; fairness remains conditional on RTT, queue, version, and flow count
Deep-buffer wired ($> 10\times$BDP)	Deep	Any	Cong.	Deep buffers can change the BBR/CUBIC share and increase queueing delay; exact fairness is configuration-dependent
WiFi 802.11n/ac (standard BBR)	Med	Low–Med	Random	Pacing, TSQ, chipset, direction and queue policy affect frame aggregation; a scalar BBR/CUBIC ratio is not transferable
LEO satellite (Starlink-class)	Varies	Varies	Mixed	High-BDP benefit may be offset by RTT variation, random loss, and path dynamics; evidence is insufficient for a fixed rating
Shallow-buf. wired ($< 1.5\times$BDP)	Shal	Low	Cong.	Bandwidth share and retransmissions require source-specific queue/RTT conditions

The WiFi row marks a known framework boundary: 802.11 frame aggregation is a protocol-layer constraint that cannot be mapped to the three physical dimensions; variants such as BBRp can partially remedy this through controlled bursts.

3.5. Analytical Assessment and Identifiability

The framework distinguishes a description of the path from a decision about performance. We formalize this distinction using a condition vector $z = (C, R_*, \{R_i\}, Q, L, v, A, n, W)$: bottleneck capacity, reference and per-flow propagation RTTs, absolute queue limit, loss process, implementation, queue policy, flow count and observation window. The physical descriptors are $\mathrm{BDP}_* = CR_*$ and $q = Q/(CR_*)$, with capacity and queue expressed in consistent bit units. These descriptors organize cases; the following analysis shows why they cannot determine bandwidth allocation by themselves.

Allocation is not identified by the in-flight identity. For two flows, let their aggregate rate be $0 < G \le C$ and let s be the second flow's share. The rates $x_1 = (1-s)G$ and $x_2 = sG$ satisfy the same aggregate-rate constraint for every $s \in [0,1]$. Their propagation-pipe amounts $I_i = x_i R_i$ also satisfy the per-flow rate–delay identities. Thus, even with C, R_1, R_2, Q fixed, the identities admit different shares; a longer propagation RTT does not algebraically force a larger share. This is an identifiability result about the equations, not a claim that all shares are equilibria of a particular controller. Selecting the realized allocation requires controller and queue dynamics or direct measurements.

Utilization and fairness are separate constraints. Define normalized receiver goodput $u = G/C$ and imbalance $d = |2s - 1|$. Substituting the two rates into Jain's standard index [143] gives

$$J = \frac{1}{2\{s^2 + (1-s)^2\}} = \frac{1}{1+d^2}. \tag{2}$$

Consequently, a fairness requirement $J \ge j_0$, where $1/2 \le j_0 \le 1$, is equivalent to

$$\frac{1 - \sqrt{j_0^{-1} - 1}}{2} \le s \le \frac{1 + \sqrt{j_0^{-1} - 1}}{2}. \tag{3}$$

For example, $j_0 = 0.95$ requires a share between 0.385 and 0.615, regardless of whether aggregate goodput fills the link. Jain's index and this algebraic transformation are established metrics, not new fairness measures. Our analytical contribution is to use their explicit feasible region to separate path descriptors, performance requirements and observable allocation in a BBR assessment.

A requirement-dependent assessment procedure. A workload specifies its requirements before protocol selection: a minimum u_0, a minimum j_0, a delay limit D_0, and a retransmission budget H_0. For each required metric, the assessment returns *meets*, *does not meet*, or *undetermined*. A missing or incompatible measurement window yields *undetermined*, rather than an inferred pass. A joint assessment meets the requirements only if every required metric meets its own limit. An observed violation yields a failure for that requirement; if there is no observed violation but some required evidence is missing, the joint result remains undetermined. A zero-goodput run has $u = 0$ and undefined Jain fairness; it cannot pass a positive utilization requirement. This procedure allows throughput and fairness evidence to be usable without claiming that they establish latency or retransmission suitability.

For repeated runs, we report both metric means and the empirical fraction satisfying the joint rate/share conditions:

$$\widehat{p}_{u,J} = \frac{1}{N} \sum_{r=1}^{N} \mathbf{1}\{u_r \ge u_0,\ J_r \ge j_0\}. \tag{4}$$

The fraction is descriptive with five repetitions per condition; it is not a population reliability guarantee. Equation 2 is applied within each run. In general, substituting an averaged share does not recover the mean Jain index. Queue bins and BDP bins do not set u_0, j_0, D_0, H_0: those limits express service requirements and must be reported separately. Section 7.5 evaluates the rate/fairness component and its transfer limitations using the archived measurements.

3.6. External Consistency Check of the Framework

To assess external consistency, we compare the conditional expectations with published measurements outside our controlled experiment matrix. The Scholz measurements provide an external comparison for the

relationships among capacity, RTT, queue depth and allocation. They were part of the evidence used to organize the framework, so the comparison is interpreted through its reported conditions.

The comparison asks whether reported behavior is compatible with three conditional expectations: throughput differences can depend on BDP and receive-buffer constraints; deep queues can affect BBR/CUBIC allocation; and changing capacity and RTT can alter the comparison. We do not require a fixed throughput multiple, a universal share threshold, or a monotonic performance ratio as BDP increases.

Scholz et al. [25] examine BBR and CUBIC under varied bottleneck bandwidths, RTTs and buffer settings in a controlled testbed. Their configuration-dependent observations motivate reporting these inputs together. This provides an external consistency comparison, not a held-out test of predictive accuracy: the study also informed the framework's organization. The controlled experiments in Section 7 provide additional measurements with explicit inputs and metric definitions.

The dimensions are related: high BDP and deep buffering can co-occur, and the observed loss process changes with load and queue management. Flow count and RTT heterogeneity also affect sharing. We therefore record implementation, AQM and competing flows alongside the three dimensions and assess throughput, delay, fairness and retransmissions separately.

4. Multi-Scenario Performance Evaluation

BBR performance depends on both network conditions and implementation. Wired backbone measurements and WLAN experiments use different link, queue, pacing and host configurations, so their throughput ratios are not interchangeable. This section examines six scenario groups while retaining those source-specific conditions. Source-specific measurements are reported in the corresponding scenario subsections and Table 7; high-dynamic routing is treated qualitatively because comparable version-by-version throughput ratios are not available.

4.1. Scenario 1: High-BDP Wide-Area Networks

High-BDP WANs can expose differences between path-model-based and loss-driven congestion control. Their performance depends on RTT, loss, receiver-window limits and implementation, rather than BDP alone. Cardwell et al. [6] report improvements for BBR in their backbone measurements. Those measurements connect the reported throughput improvements to their path and host conditions.

Sustaining a target rate also requires a receive window large enough for the corresponding rate–RTT product. The original paper's controlled experiment shows that once the receive window is limited, BBR's algorithmic advantage is masked by the receiver bottleneck and throughput drops to levels comparable to CUBIC [6].

4.2. Scenario 2: Shallow-Buffer Wired Networks

Shallow-buffer comparisons must report absolute queue size, capacity, reference RTT, implementation and flow count together. A queue size in kilobytes cannot be compared across studies without those conditions. The cited Cao and Jaeger studies use different configurations, so their observations are not combined into a single queue-sweep table here [23, 24].

The inspected Jaeger author manuscript reports a configuration-dependent relationship between buffer size and BBR/CUBIC shares. Its Figure 10 uses both common-RTT and asymmetric-RTT experiments; those observations do not establish a universal retransmission multiplier or a fixed suitability rating. When interpreting competition, distinguish the TCP recovery logic from the congestion controller's response to loss, and report receiver goodput separately from retransmission counts. Applications with different latency, delivery and retransmission requirements may assess the same configuration differently.

Cao et al. [23] report a sharp goodput decline near 20% configured random loss in their 100-Mbps, 25-ms, 10-MB-buffer experiment. This source-specific loss sweep does not establish that BBR is generally unaffected below 20% loss, or transfer that boundary to later versions.

4.3. Scenario 3: Stable Broadband Multi-Flow (BBRv3 Benchmark)

Gomez et al. [26] evaluate BBRv3 using distinct emulated and hardware configurations. The reported comparisons vary with loss, RTT, buffer size, competing flows and queue management; they should not be summarized as a fixed throughput multiplier over CUBIC.

In the emulated loss sweep (Section 5.4, Figure 7), the study uses a 1-Gbps bottleneck, a reported 20-ms propagation delay, a BDP-sized buffer, and ten 120-s runs per loss rate. BBRv2 and BBRv3 throughput declines noticeably above 0.1% loss and reaches about 60% of its maximum at 0.5% loss. The reported throughput remains above 20% until loss exceeds 1%; this does not imply full throughput at 1% loss. These approximate source-reported values describe that configuration rather than universal loss thresholds.

In the separate low-RTT hardware experiment, two VMs communicate over a 100-Gbps link without an emulated bottleneck or artificial loss. Each configuration runs for 120 s and is repeated ten times, with parallel-stream counts and MTUs varied. Figure 13 reports higher CUBIC throughput than BBRv3 in the tested configurations. The high-capacity, low-delay hardware experiment emphasizes raw transfer rate, while the loss and queue experiments assess different aspects of congestion response and sharing. Their configurations define the scope of each comparison.

Figure 5 compares two same-version BBR flows with reported 20-ms and 100-ms delays. For BBRv3, replacing Tail Drop with FQ-CoDel yields near-equal throughput and a Jain index close to one across the plotted buffer sizes. FQ-CoDel combines per-flow scheduling with CoDel, so this comparison does not isolate the effect of CoDel alone or establish inter-protocol fairness with CUBIC. The source uses both RTT and propagation-delay terminology for these delay values.

The FQ-CoDel comparison identifies queue management as an assessment input alongside buffer depth and RTT heterogeneity. Its reported fairness values are interpreted with the corresponding configuration and metric definition.

4.4. Scenario 4: High-Loss Wireless Networks (WiFi/Cellular)

The core argument for wireless scenarios is that BBR's suitability is not determined by the wireless channel's random loss rate alone, but by the *protocol-layer characteristics* of the loss source—cellular and WiFi impose fundamentally different constraints on BBR due to different physical layer mechanisms. In real LTE cellular networks, BBRv1 downlink average throughput outperforms CUBIC with lower RTT [144]; in IEEE 802.11n/ac WiFi results depend on the sender location, wireless chipset, bottleneck position, queue discipline, and TCP Small Queues (TSQ) settings. Grazia et al. [139] study an Arch Linux 4.19 physical testbed with IEEE 802.11n/ac links and distinguish upload from download experiments. The study uses Flent and repeats each experiment ten times. Each test lasts 40 s, with TCP traffic active during the central 30 s and ICMP probes before and after that interval. The reported box summaries show medians and the 10th/90th percentiles across repetitions. Single-direction tests and the mixed four-upload/four-download RRUL workload are evaluated separately. Their BBRp modification makes the ProbeBW probing gain configurable; the tested gain of 1.5 replaces the original 1.25. It is a pacing modification, not evidence that all BBR variants require the same change.

The authors show that pacing and local queue constraints can limit frame aggregation. In their download comparison, the sender is on the wired side, so sender-local TSQ is not varied as the wireless bottleneck parameter. Loss-based CUBIC and NewReno approach the tested wireless capacity at a higher RTT, while BBRp improves the throughput–latency tradeoff relative to the standard BBR implementation. The direction, chipset, queue policy and TSQ configuration identify the conditions under which this throughput–latency tradeoff was measured.

4.5. Scenario 5: High-Dynamic Routing (SDN Rerouting/Link Fluctuation)

Path changes can alter capacity and delay on different time scales. The cyclic behavior examined by Miyazawa et al. [145, 146] shows why adaptation and competition should be evaluated over time, rather than inferred from a static buffer category. A maximum delivery-rate filter can retain earlier high samples after a capacity decrease, so its presence alone does not prove sub-RTT detection or one-RTT convergence. ProbeRTT entry also depends on sample freshness and controller state, rather than an unconditional ten-second schedule. For this scenario, capacity-change frequency and recovery time are additional assessment inputs alongside the static path descriptors.

4.6. Scenario 6: LEO Satellite Networks (Starlink-Class)

LEO satellite links combine changing capacity, delay and loss. Their BDP must be computed from a reported capacity and reference RTT for the particular direction and observation period, rather than assigned

a fixed "high" category. Measurements on Starlink and other satellite links provide different workloads and path conditions [12, 147]; their throughput comparisons should not be interpreted as a universal BBR advantage.

Delay spikes and handover-related changes motivate evaluation of estimator freshness and adaptation. Window adjustment and handover-aware QUIC designs are relevant approaches [13, 148], but a Satellite comparisons retain terminal, direction, capacity variation and RTT series so that reported gains can be interpreted under their deployment conditions. These inputs determine the BDP classification and suitability assessment.

Table 7
Conditional assessment across six scenarios (with three-dimensional coordinates and deployment criteria).

Scenario	3-D coords	Key evidence	Deployment criterion	Risk and reference
High-BDP WAN	M/H/ Cong	Source-specific throughput observations; no transferable numerical ratio	recv. buffer $\geq$ BDP	Check receive-window limitation [6]
Shallow-buffer wired	Sh/L/ Cong	Source-specific share/retransmission observations; no fixed multiplier	Assess retx budget first	Check queue-dependent coexistence [23, 24]
Broadband multi-flow	M/M/ Cong	No fixed throughput gain; source-specific buffer/RTT/AQM conditions	Combine AQM + buffer config	Verify fairness across queues [26]
High-loss wireless	M/L–M/Rnd	Cellular/WLAN results require separate source conditions; pacing and aggregation interact	Check frame aggr. / AQM	Verify the tested WiFi configuration [139, 144]
High-dynamic routing	Var./Mix	Filter memory and path changes affect adaptation; no fixed convergence time	Monitor switch frequency	Check transient adaptation [145]
LEO satellite	Var./Mix	Source-specific satellite comparisons; capacity and delay vary with path and terminal	Monitor RTT spikes + terminal	BBRv3 evidence is setup-specific [12, 13]

5. Structural Analysis of Fairness Challenges

BBRv3's intra-protocol fairness remains sensitive to buffer depth and RTT heterogeneity. In the multi-RTT simulation of Piotrowska [18], the reported Jain index varies across BBR versions and buffer settings; for the nine-path access-link scenario, all link rates are 100 Mbps and RTTs range from 8 to 204 ms. The study's Table 8 reports BBRv3 Jain indices of 0.70, 0.62, 0.81, and 0.86 at buffer sizes of 0.5, 1, 2, and 10 times the BDP, respectively, where the BDP is referenced to the mean RTT of 100 ms. These results are conditional on that simulation's topology and RTT distribution. These measured differences motivate reporting buffer settings alongside Jain fairness and per-flow rates. The identity BDP = BtlBw $\times$ RTprop determines an in-flight amount associated with a measured operating point, but does not by itself determine bandwidth allocation. Observed shares also depend on probing dynamics, queue depth, version, queue management, and flow count.

This section distinguishes two comparisons. Intra-protocol fairness concerns bandwidth sharing among flows using the same controller, including flows with different RTTs. Inter-protocol fairness concerns coexistence with a different controller such as CUBIC. Both depend on the measured rates and the network configuration. Figure 5 summarizes the assessment inputs; the version comparison below summarizes reported mechanisms and their limits across versions.

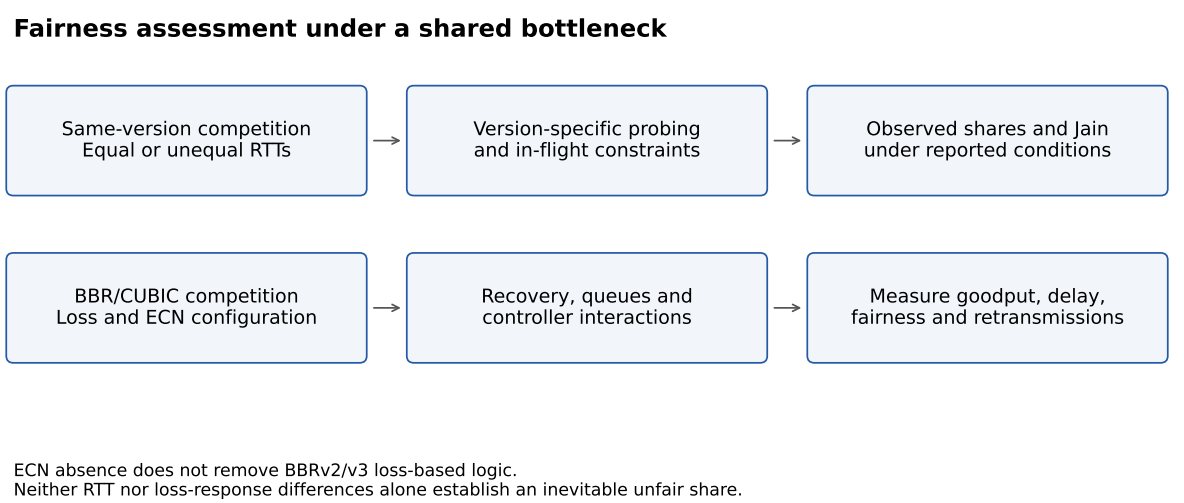


Figure 5: Assessment of same-version and BBR/CUBIC competition. Controller responses interact with queueing, recovery and the experiment configuration; fairness outcomes require measured per-flow rates. The original figure was generated with GPT Image 2.0 and manually checked by the authors; this replacement was drawn with Python/matplotlib using AI-assisted code preparation.

5.1. Formal Fairness Metrics

Quantitative fairness assessment requires explicit metrics. The Jain Fairness Index (JFI) proposed by Jain et al. [143] is the primary tool for measuring multi-flow bandwidth allocation in this paper:

$$J(\mathbf{x}) = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \cdot \sum_{i=1}^{n} x_i^2} \tag{5}$$

where $\mathbf{x}$ is the throughput vector of n flows, $J = 1$ indicates perfect fairness, and $J = 1/n$ indicates single-flow monopoly. In BBR analysis, JFI must be combined with per-flow absolute throughput: when the number of flows is large, the index may mask starvation of low-bandwidth flows [20].

5.2. Intra-Protocol Fairness under Heterogeneous RTTs

When multiple BBR flows compete at the same bottleneck, RTT heterogeneity can produce unequal throughput shares, often disadvantaging short-RTT flows under the conditions reported in prior measurements—a phenomenon commonly called "RTT bias." Hock et al. [22] first systematically reported this problem: at 0.8×BDP buffer, short-RTT flows have a slight advantage; at 8×BDP, long-RTT flow throughput can be several times that of short-RTT flows. Tao et al. [149] analyze RTT fairness using a queueing model. Any predicted throughput ratio or fairness threshold depends on that model's assumptions and must be checked against the experiment's queue, probing and flow conditions. It is not interpreted here as a rule that an RTT ratio above two inevitably produces a particular Jain index.

The per-flow BDP calculation relates two measured quantities:

$$\mathrm{BDP}_i = \mathrm{BtlBw}_i \times \mathrm{RTprop}_i \tag{6}$$

The longer-RTT flow has a larger RTprop_i, so the in-flight amount associated with a given measured delivery rate is larger. This relationship does not by itself determine the flow's bandwidth share: probing duration, queue depth, max-filter dynamics, version, queue management, and competing flow count also matter. Within BBR's existing "BtlBw×RTprop" measurement model, adjusting gain parameters may change the observed bias under particular conditions, but the equation alone is not a proof of inevitable long-RTT dominance. A stronger mitigation may require changes to path-delay estimation or additional network-side signals such as scalable ECN.

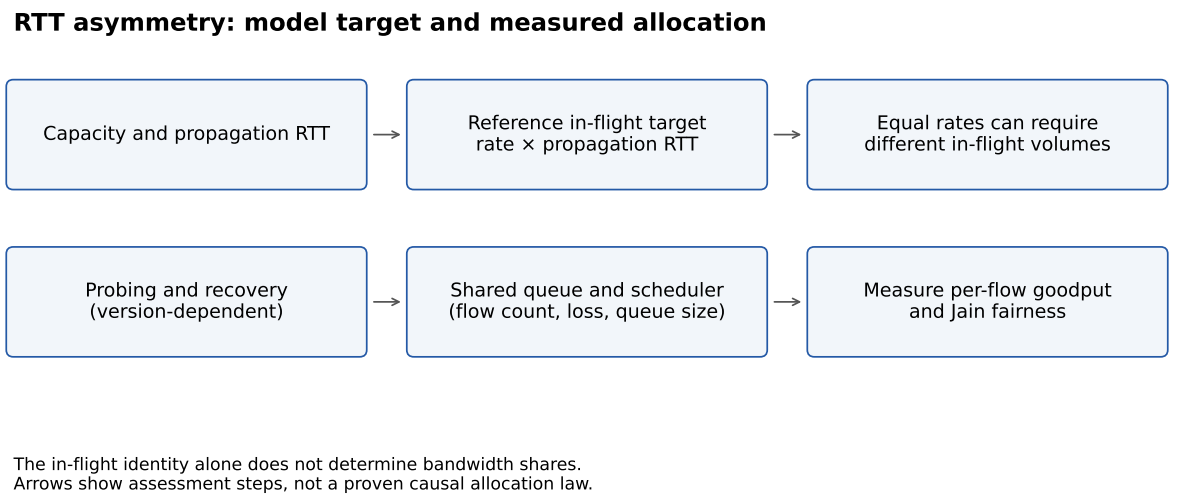


Figure 6: RTT asymmetry separates the reference in-flight target from bandwidth allocation. Equal rates may require different in-flight volumes. Probing and shared-queue interactions must be evaluated through per-flow goodput and Jain fairness; no RTT-proportional share law is assumed. The original figure was generated with GPT Image 2.0 and manually checked by the authors; this replacement was drawn with Python/matplotlib using AI-assisted code preparation.

BBRv2 attempts to mitigate intra-protocol unfairness by limiting inflight when ECN marking or loss feedback exceeds a threshold [7]. Without ECN marking, the ECN-derived component is unavailable, but loss-driven updates and the associated state logic remain. BBRv3 further changes the ProbeBW Down pacing behavior [8]. In Piotrowska's multi-RTT evaluation, the BBRv3 Jain index varies across buffer settings and is not uniformly higher than BBRv2; the exact values are conditional on that study's topology and RTT distribution. The measured Jain values make version and queue configuration joint inputs to the fairness assessment.

5.3. Inter-Protocol Unfairness: Coexistence with CUBIC

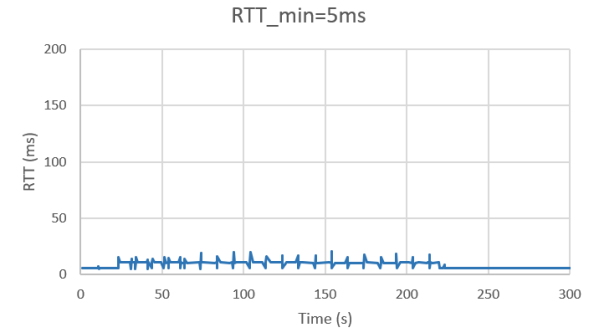
When BBR and CUBIC coexist, buffer size, RTT configuration, and flow count can affect their bandwidth shares. In the emulation study of Jaeger et al. [24], Figure 10a varies the buffer with both flows at 50 ms RTT, Figure 10b varies their common RTT at a 2.3-BDP buffer, and Figure 10c holds one flow at 50 ms while varying the other. The reported fairness measure is the authors' throughput-based F_{tp}; it should not be interchanged with Jain's index.

For that tested implementation, the authors report persistent BBR-induced loss at buffer sizes up to about 1.5 BDP, approximately balanced shares near the intermediate buffer range, and increasing CUBIC share as the buffer grows further. These observations support a configuration-specific buffer effect. They do not establish a universal 1.5- or 3-BDP threshold for BBRv2, BBRv3, other queue policies, or arbitrary flow populations. The cited survey literature provides context for these coexistence problems, but a quantitative comparison requires the original study, implementation, flow configuration, and metric definition to be identified. Zeynali et al. [21] further observed more extreme results on real Internet paths: a single BBRv3 flow can occupy over 99% of link bandwidth (vs. 5 CUBIC flows) when ECN is enabled, and disabling ECN only marginally reduces unfairness [21].

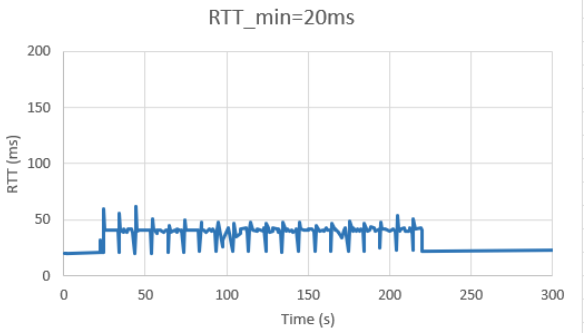
5.4. Existing Remedies and Their Limits

For **intra-protocol fairness**, BBQ [150][1] reduces probe cycles for long-RTT flows; DA-BBR [151] introduces a correction factor for per-flow BDP; FaiRTT [152] achieves a Jain index of 0.98 in ns-3

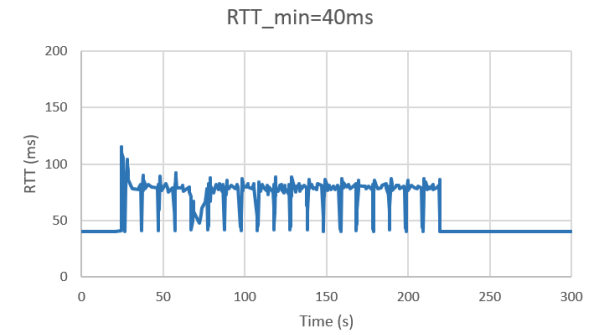
[1]Published as an arXiv preprint (arXiv:1706.09115); formal publication not confirmed at time of review. Cross-validate with Tao et al. [149].



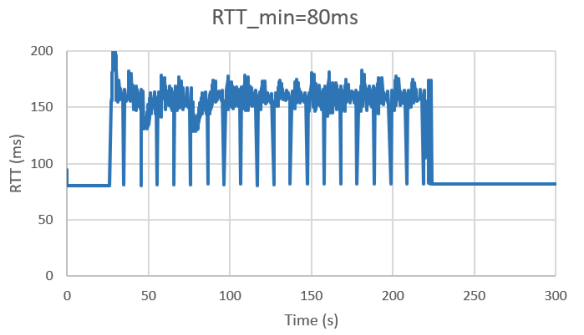
(a) Longer-flow $\mathrm{RTT}_{\min} = 5\,\mathrm{ms}$ ($\Delta\mathrm{RTT} = 0\,\mathrm{ms}$, baseline)



(b) Longer-flow $\mathrm{RTT}_{\min} = 20\,\mathrm{ms}$ ($\Delta\mathrm{RTT} = 15\,\mathrm{ms}$)



(c) Longer-flow $\mathrm{RTT}_{\min} = 40\,\mathrm{ms}$ ($\Delta\mathrm{RTT} = 35\,\mathrm{ms}$)



(d) Longer-flow $\mathrm{RTT}_{\min} = 80\,\mathrm{ms}$ ($\Delta\mathrm{RTT} = 75\,\mathrm{ms}$)

Figure 7: RTT of the shorter-RTT flow under BBR two-flow competition. The vertical axes show RTT, including propagation and queueing components. The shorter flow has $\mathrm{RTT}_{\min}$ fixed at 5 ms; the longer flow's $\mathrm{RTT}_{\min}$ varies from 5 ms to 80 ms, giving RTT differences of 0, 15, 35, and 75 ms respectively. Redrawn based on data from Hock et al. [22].

simulation, but robustness in real wireless networks is yet to be verified. These approaches intervene through probing, operating-point estimation and window control, offering distinct mechanisms for changing measured allocation. Their comparisons should retain the tested implementation and network conditions.

For **inter-protocol fairness**, Modest-BBR [153] scales the congestion window after loss; BBR-CWS [154] dynamically adjusts the congestion window during loss recovery, but lacks response to non-loss congestion signals (ECN). The IETF L4S standard [15] represents another technical path: introducing a dual-queue structure and scalable ECN congestion signals at the network side to support both low latency and high throughput—complementary to, rather than a replacement for, end-side algorithm iteration.

Table 8 connects version-specific mechanisms to reported sharing behavior. Estimator design, probing dynamics, loss responses and queue management should be evaluated jointly with per-flow rates and explicit measurement conditions.

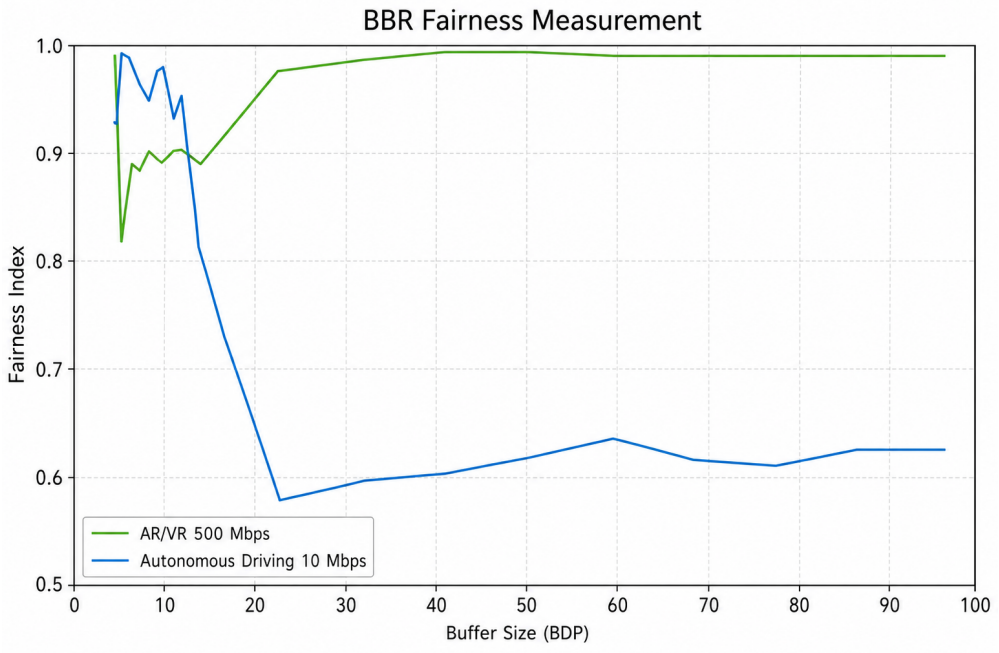


Figure 8: Fairness variation of BBRv3 and CUBIC coexistence as a function of bottleneck buffer size [20].

Table 8
Summary of fairness improvement trajectory across BBRv1/v2/v3.

Version	Intra-protocol	Inter-protocol	Key improvement	Assessment considerations
BBRv1	RTT-dependent shares under reported queue/probing conditions	BBR/CUBIC shares vary with buffer, RTT and flow count	—	Cwnd-derived inflight limit remains; loss response differs from v2/v3
BBRv2	Conditional improvement under the reported configurations	Improved under some ECN-enabled configurations	ECN triggers $\mathrm{inflight_hi}$ constraint	ECN-derived feedback is unavailable on non-ECN paths; loss-driven logic remains
BBRv3	**Fairness remains configuration-dependent with deep queues**	No universal improvement over v2 established	Version-specific probing and loss-response changes	Reported fairness limitations require their tested implementation and queue conditions

6. Engineering Obstacles and AI-Enhanced Directions

6.1. Engineering Deployment and Structural Obstacles

Deployment comparisons evaluate parameter tuning, estimator changes and network-side mechanisms under the same stated performance requirements. The relevant outcomes are per-flow rates, delay, retransmissions and operating cost.

BBR has moved from experimental scheme to large-scale engineering deployment since 2016: Google B4 and YouTube/CDN public observations show it can improve utilization, reduce latency, and alleviate Bufferbloat [6]; the Linux kernel and IETF drafts drive v1–v3 continuous evolution [6, 7, 19, 20]; and Starlink-class link measurements show BBR has potential advantages on LEO satellite paths, but with strong dependence on terminal, tunnel, and switching conditions [12, 13].

Deployment assessment should examine feedback availability, wireless measurement conditions and application requirements separately.

A deployment consideration is the availability of explicit congestion feedback. ECN marking requires support at the endpoints and along the path, and L4S additionally depends on its queueing and endpoint requirements [14–17]. Without marking, the ECN-derived component is unavailable, but BBRv2/v3 retain loss-driven updates and version-specific state logic. Performance under that configuration must be measured; absence of ECN does not establish reversion to BBRv1 or a fixed loss of fairness.

The second obstacle is **multiple incompatibilities in wireless environments**, each with a different root cause, making a single fix insufficient. An underestimated RTprop can reduce the model-derived in-flight target and delivered throughput [20]; pacing and local queue constraints can limit WiFi frame aggregation in the tested 802.11n/ac settings (see §4.4); and the time-scale mismatch between wireless/LEO link variation and BBR's multi-RTT probing windows can cause measurement model estimates to lag channel state changes [13, 124, 148].

The third obstacle is the **absence of heterogeneous flow QoS guarantees**. Bandwidth fairness alone does not establish latency or reliability guarantees for heterogeneous applications. Claims about URLLC suitability require application-level deadlines, loss budgets and scheduling conditions; a bulk two-flow result does not supply those guarantees.

6.2. Classification Evaluation of AI-Enhanced Approaches

Evidence on learning-assisted control should distinguish the action space and evaluation environment. Xiao et al. [155] learn congestion-window actions in ns-3 using a goodput–delay utility difference. In their dynamic simulation, Table 2 reports a mean Jain index of 0.8058 and a 95% confidence interval of [0.7233, 0.8882]. The baseline intervals overlap, and the table alone does not establish statistically superior fairness. The authors also report that performance is best near the training parameters. These observations motivate explicit tests of transfer and inference cost. The comparison concerns the reported baseline set, which excludes BBR.

Learning-based methods should be assessed by their observations, actions, objectives and evaluation settings. Learning a local BBR parameter differs from replacing the controller policy or coordinating several flows.

One research direction is learning-assisted local estimation and decision-making: Ketabi et al. [163] use DRL to adjust pacing_gain and cwnd_gain; OI-BBR [159] uses a decision tree to learn "when to yield to CUBIC." Local parameter changes preserve more of the underlying controller than a full policy replacement. Potential uses include RTprop estimation calibration, WiFi frame-aggregation identification and ProbeRTT timing; their applicability depends on the available signals, action space and evaluation conditions. Local parameter learning is evaluated against fixed heuristics and broader policy changes using per-flow rates, delay and operating cost.

Policy-level learning schemes explore a wider action space than local parameter adjustment [156–158, 161, 162]. Their deployment assessment should report inference time, training cost, controller stability and performance outside the training distribution. Network-level methods such as GraphCC and LLM-CC require a separate assessment of coordination scope and decision frequency [164, 165]. Deployment evaluation pairs control performance with measured inference latency at the intended decision frequency.

Table 9
Comparison of representative AI-assisted congestion control approaches (★: BBR-specific works). Remy [156]; PCC Vivace [157]; Aurora [158]; OI-BBR [159]; Symbolic-PCC [160]; MARLIN [161]; Mutant [162]; RL-BBR [163].

Method	AI paradigm	Ctrl level	Testbed	Interp.	Key contribution	Evaluation scope and deployment requirements
Remy	Offline opt.	Policy	Sim	Low	Auto-generated CC policy	Requires prior distribution
PCC Vivace	Online learning	Policy	R+S	Low	Utility-driven control	Slow convergence
Aurora	DRL	Policy	Sim	Low	Multi-scenario generalization	Sim2Real gap
★ OI-BBR	Decision tree	Assist	Mininet	**High**	Interpretable rule learning	Discrete decisions only
Symbolic-PCC	DRL+sym. distill.	Param	Sim+tb	Med	Symbolic rules from NN	Limited expressiveness
MARLIN	Max-ent. RL	Policy	R+S	Low	Mitigates Sim2Real gap	High complexity
★ Mutant	Online RL+imit.	Policy	Real	Low	No offline training	Stability concerns
★ RL-BBR	DRL	Param	DC	Low	DC throughput optimization	Narrow scope

6.3. Core Open Challenges

The reviewed evaluations leave several questions for deployment. First, throughput–delay utility is not equivalent to an explicit fairness or service-priority objective. TCP-Drinc reports Jain fairness, but its baseline intervals overlap and its evaluation does not include BBR. A study's reported metrics must therefore be distinguished from the objectives used to train its policy; we make no claim that all learning methods optimize only one objective or that none can address fairness.

Second, interpretability should be evaluated alongside performance and operational safeguards. A decision tree exposes rules more directly than a neural policy, but neither representation alone proves stability or justifies deployment. Third, an evaluation in ns-3 or Mininet is limited by the topology, traffic, channel and timing models actually configured, not by an inherent inability of those tools to model dynamic paths. Transfer claims need tests outside the training conditions, including heterogeneous traffic and path changes [166]. Finally, wireless and LEO evaluations should test changes in capacity, delay and loss, as well as adaptation time and retained performance [13, 124]. These are evaluation questions, not a proof of a universal ceiling on learning-assisted control.

6.4. Technical Roadmap

The reviewed evidence motivates two complementary research directions, without excluding other designs. The near-term path is a "model-driven + limited data-driven" hybrid framework: keep the BBR state machine auditable, and limit learning modules to local sub-problems such as RTprop calibration, WiFi frame aggregation detection, or ProbeRTT timing adjustment. A complementary direction is network-side congestion feedback. L4S dual queues and scalable ECN (RFC 9330 [15]) require compatible endpoint behavior. Whether a BBR implementation meets those requirements and improves fairness must be evaluated; L4S support alone does not establish that outcome. The former improves end-side measurement accuracy, the latter improves network-side signal quality; they are complementary, not alternatives.

7. Controlled Experiments

7.1. Implementation identity and experimental design

We use E1 to examine bandwidth sharing between two flows of the same implementation as RTT asymmetry and queue depth change, and E2 to examine coexistence with CUBIC. E3 extends the equal-RTT comparison across capacity and propagation delay. An artificial-loss extension separates configured access-link loss from bottleneck congestion. These tests evaluate the framework's assessment dimensions under stated conditions, Each experiment addresses a separate assessment question.

The Linux emulation uses Mininet 2.3.0, Open vSwitch bridges, iperf3 3.9 and iproute2 5.15.0. Two sender hosts connect through switches s_1 and s_2 to one receiver: $h_1, h_2 \rightarrow s_1 \rightarrow s_2 \rightarrow h_3$. The s_1–s_2 link is the bottleneck, configured with a token-bucket rate limiter and child netem FIFO packet limit, without AQM or

ECN. The receiver access link supplies the base propagation delay; additional delay on the second sender's access link creates RTT asymmetry. Each sender opens one TCP connection to a separate receiver port (5201 or 5202), with one-second iperf3 reporting intervals. The topology is rebuilt between runs. The MTU is 1,500 bytes. In E1/E2, the reference BDP is $100\,\text{Mbps} \times 20\,\text{ms}/8 = 250,000$ bytes, using the first flow's configured propagation RTT, including when the second RTT differs. Queue targets of 125,000, 500,000 and 2,500,000 bytes are rounded up to 84, 334 and 1,667 MTU-sized packets, respectively.

The archived experiments comprise 340 two-flow runs: 60 each for the Ubuntu 5.15.0-181-generic in-tree baseline, the separately loaded BBRv1 module, BBR2, and BBRv3; 80 E3 runs; and 20 artificial-loss runs. The manifests identify BBRv3 by the 6.13.7-bbr3 kernel, source marker BBR_VERSION=3, and runtime name `bbr`. The BBRv1 module registers as `bbr1` and uses a recorded compatibility patch for an unexported TSO helper. BBR2 uses a separate 5.13.12-bbr2alpha kernel and runtime name `bbr2`. The in-tree baseline remains a distinct implementation label. Different kernel environments and the module patch limit causal attribution of performance differences to algorithm version alone.

The E1/E2 matrix uses a 100-Mbps bottleneck, a 20-ms first-flow RTT, second-flow RTTs of 20, 40, or 80 ms for the same-algorithm comparison, and equal-RTT BBR/CUBIC competition. Queue targets are 0.5, 2, or 10 times the reference BDP. Each condition has five 180-s repetitions, with ECN disabled. Receiver-side iperf3 summaries provide the archived full-run goodput. Jain fairness and flow shares are derived from those per-flow rates. For each run, with receiver rates x_1, x_2, we compute $G = x_1 + x_2$, $J = (x_1 + x_2)^2/[2(x_1^2 + x_2^2)]$ and $S_2 = x_2/(x_1 + x_2)$. Reported Jain indices and shares are means of these run-level quantities, rather than values calculated from mean rates. In E2, flow 1 is BBR and flow 2 is CUBIC, so the mean BBR share is $1 - \overline{S_2}$. Receiver goodput excludes protocol headers; therefore it need not equal the configured link rate. The supplement supplies per-run values and variability; five repetitions support descriptive comparisons, not population-wide significance claims.

The configuration declares a 30-s warmup; the primary archived analysis uses the complete receiver summary. A separate sensitivity analysis uses complete receiver intervals within the configured 30–180-s window, without estimating bytes in boundary intervals. Across 68 condition summaries, the largest absolute changes in mean total goodput, Jain index, and second-flow share are 1.313 Mbps, 0.02949, and 0.01678, respectively. The separate window analysis quantifies the sensitivity of these summaries.

Loaded RTT measurements are sequential early ping samples from the two senders. Queue counters were sampled before the transfers completed. The RTT and queue observations use different windows from the receiver goodput summary; their values are reported as complementary measurements.

7.2. Conditioned cross-version comparison

Across the tested 100-Mbps matrix, equal-RTT total goodput remains close to the bottleneck for all four implementations, while flow shares and Jain fairness vary with RTT asymmetry, queue setting, and implementation. Under the 40- and 80-ms second-flow conditions, the observed long-RTT share differs across implementations. E1 quantifies allocation under RTT asymmetry; E2 evaluates coexistence with CUBIC at equal RTTs in the documented implementation environments.

For the no-ECN BBR/CUBIC comparison, total goodput is similar across the four implementation environments, while the BBR/CUBIC share and Jain index remain condition dependent. The experimental supplement contains the per-run rates, condition summaries and plotting scripts.

Table 10 shows that similar total goodput does not imply similar bandwidth allocation. Total rates are 95.58–95.59 Mbps in this matrix, while the BBR share is 0.143–0.327. BBR2 obtains shares of 0.274–0.327 and Jain indices of 0.828–0.886; the BBRv1 module and BBRv3 obtain smaller shares in these particular runs. The implementation environments and no-ECN configuration identify the scope of these comparisons.

For unequal-RTT E1 cases, the second flow receives substantially more than half the goodput in the tested matrix (Figure 9). The size of the imbalance varies with implementation and queue setting: across the 24 unequal-RTT conditions, the mean long-RTT share ranges from 0.769 to 0.969, and mean Jain fairness from 0.532 to 0.776. These measured shares make allocation an explicit assessment outcome, alongside the aggregate rate and the reported path conditions.

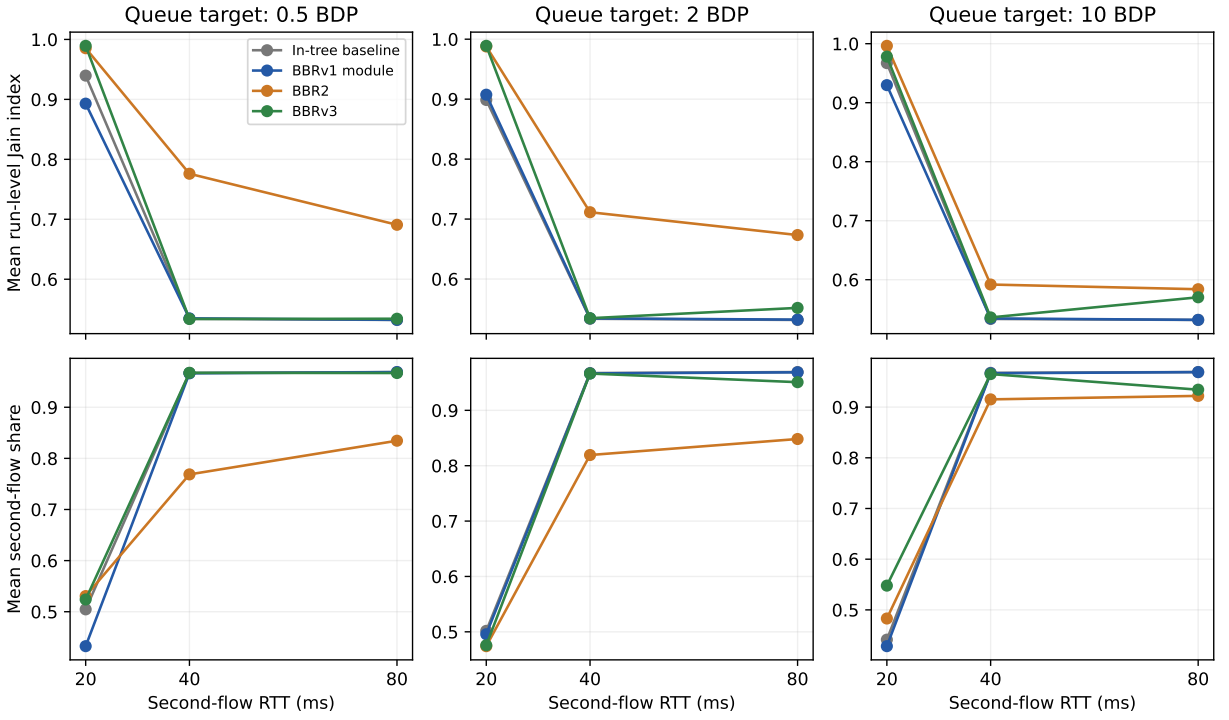


Figure 9: E1 two-flow comparison at 100 Mbps, with first-flow RTT 20 ms and ECN disabled. Curves show means of five run-level Jain indices and second-flow shares, not indices computed from averaged rates. Queue targets use the reference BDP.

7.3. Additional comparison under equal RTT

The additional experiment matrix contains eight network cases at 50 or 200 Mbps, 10 or 60 ms RTT, and two frozen buffer labels. For the 50-Mbps cases, the configured absolute buffers correspond to 4 and 16 times the physical reference BDP, although the frozen file labels them 1 and 4. For the 200-Mbps cases, those labels are consistent with the capacity-times-reference-RTT BDP. The original predictions are retained; the affected cases do not validate the intended buffer bins. We ran two BBRv3 flows and two CUBIC flows separately for five independent repetitions per case (80 runs in total). ECN is disabled in all cases.

Table 11 reports mean total goodput and Jain fairness. E3 extends the rate/fairness assessment across two capacities and two RTTs. Mean BBRv3 Jain indices range from 0.995 to 1.000, compared with 0.917 to 0.987 for CUBIC; the table reports the corresponding goodput values together, making both utilization and sharing visible. The queue snapshots do not cover the complete measurement window; The receiver summary, sender retransmissions and queue snapshot provide separate indicators for the documented operating conditions. The archived goodput and Jain values use full-run receiver summaries. A separate post-warmup interval sensitivity analysis is retained; the table reports the complete receiver summary; window sensitivity is provided separately.

7.4. Goodput under artificial access-link loss

The artificial-loss extension compares BBRv3 and CUBIC at configured loss probabilities of zero and 0.1% on a separate access link. Each run uses two flows of the same algorithm, a 100-Mbps bottleneck, equal 20-ms RTTs and a 500,000-byte queue target (2 reference BDP), with five runs per algorithm and loss setting. Archived full-run mean total goodput changes from 93.87 to 94.25 Mbps for BBRv3 and from 95.58 to 81.06 Mbps for CUBIC. The small BBRv3 increase is a descriptive observation and is not evidence that loss improves performance. The loss generator's seed was not exposed in the retained configuration; this setup does not represent correlated wireless or satellite errors.

Table 10
E2 competition with CUBIC at 100 Mbps and equal 20-ms RTTs, with ECN disabled. Each cell summarizes five full-run receiver measurements. Share denotes the BBR flow; Jain is the mean of per-run indices. Different implementation environments limit causal version attribution.

Implementation	Q/BDP	Total (Mbps)	BBR share	Jain
In-tree baseline	0.5	95.59	0.155	0.677
In-tree baseline	2	95.59	0.153	0.674
In-tree baseline	10	95.58	0.165	0.690
BBRv1 module	0.5	95.59	0.144	0.663
BBRv1 module	2	95.59	0.143	0.663
BBRv1 module	10	95.59	0.149	0.669
BBR2	0.5	95.58	0.278	0.835
BBR2	2	95.58	0.327	0.886
BBR2	10	95.58	0.274	0.828
BBRv3	0.5	95.58	0.153	0.675
BBRv3	2	95.58	0.145	0.664
BBRv3	10	95.59	0.164	0.689

Table 11
Descriptive E3 full-run means. Goodput (Mbps) and Jain cells report BBRv3/CUBIC. C is Mbps and RTT is ms. Buffer uses decimal kB; actual target Q/BDP is computed from capacity and reference RTT, before packet-limit rounding. Frozen labels are preserved in the input audit.

Case	C	RTT	Buffer (kB)	Actual Q/BDP	Goodput	Jain
E3_001	50	10	250	4	47.1/47.8	1.000/0.969
E3_002	50	10	1000	16	47.2/47.8	1.000/0.917
E3_003	50	60	1500	4	46.9/47.7	1.000/0.928
E3_004	50	60	6000	16	46.8/47.7	1.000/0.987
E3_005	200	10	250	1	188.2/191.2	0.995/0.963
E3_006	200	10	1000	4	188.0/191.2	0.999/0.955
E3_007	200	60	1500	1	187.0/190.8	0.997/0.963
E3_008	200	60	6000	4	187.1/190.8	0.996/0.938

The additional equal-RTT cases extend the capacity and RTT settings of the main matrix. Their physical queue sizes are reported explicitly in Table 11. Because four cases differ from the intended buffer labels, we use these results as an additional empirical comparison, rather than as a validation of the original buffer-category predictions. The experiments evaluate goodput and bandwidth sharing under controlled conditions; they do not establish universal suitability thresholds or isolate the causal effect of an individual controller mechanism.

7.5. Quantitative Transfer Evaluation of the Framework

We supplement the scenario comparisons with a quantitative test of whether an equal-RTT calibration transfers to different capacities and delays. This analysis was designed during revision after the archived results were available. Calibration and evaluation runs are disjoint, but the test is retrospective; its design and data split are documented in the supplement. The analysis script retains all inputs and reports the constant-baseline comparison, including the contribution of queue conditioning.

Calibration uses only the 15 BBRv3 E1 runs with equal 20-ms RTTs at 100 Mbps, five at each queue target $q \in \{0.5, 2, 10\}$. At each target, we estimate mean u and mean J. For a target queue ratio between knots, we linearly interpolate the two means in $\log q$; outside the range, we extend the adjacent segment and restrict predictions to the metric's feasible interval. The comparator predicts the overall calibration mean, independent of queue size. Neither model fits an E3 outcome. Capacity and RTT effects cannot be estimated from this single-capacity, equal-RTT calibration; transferring normalized rates therefore tests an explicit assumption that the evaluation tests.

The primary transfer comparison uses the four 200-Mbps E3 cases, where physical queue ratios agree with the frozen input labels: RTTs of 10 or 60 ms and $q \in \{1, 4\}$. The four 50-Mbps cases are a separate comparison using their corrected physical ratios of 4 and 16. Errors are computed against each condition's five-run mean, for 20 BBRv3 runs in each group. The 40 CUBIC runs remain a performance comparator and are not calibration data for the BBRv3 predictor.

Table 12 separates modest transfer error from predictive added value. In the four input-matched cases, the queue-conditioned Jain estimate has slightly smaller MAE than the constant predictor, but normalized-goodput MAE is the same to the displayed precision. In the corrected 50-Mbps cases, Jain MAE is 0.02027 for queue interpolation and 0.01444 for the constant estimate. This comparison identifies the dependence of transfer estimates on their conditioning variables. The largest Jain error is about 0.025; reporting only high aggregate utilization would conceal this assessment limitation. These results quantify the framework's transfer boundary without claiming that E3 validates the original queue thresholds.

We also instantiate the analytical assessment in Section 3.5 with illustrative requirements $u_0 = 0.90$ and $j_0 = 0.95$. These values were selected for retrospective illustration, not frozen as experimental acceptance thresholds. All 40 BBRv3 and all 40 CUBIC E3 runs meet the utilization requirement, while 40 BBRv3 and 27 CUBIC runs meet both requirements. The result is restricted to equal-RTT same-algorithm competition; E2 provides the separate coexistence assessment. The supplement reports run-level decisions and a threshold sensitivity grid. Receiver-goodput and early-ping windows are reported separately. Sender retransmission totals and queue snapshots provide complementary indicators for the documented operating conditions.

The higher utilization requirement in Table 13 reverses which algorithm meets the joint requirement more often: no BBRv3 run reaches $u = 0.95$, whereas all CUBIC runs do. Tightening fairness instead reduces both algorithms' pass counts. This measured throughput–allocation tradeoff makes the service preference explicit under equal RTTs. The full threshold grid and per-condition variability are provided in the supplement.

The frozen F07 predictions are retained separately from this new analysis. They specify qualitative "near-capacity" or "near-one" outcomes without numerical acceptance margins, and request medians rather than the reported full-run means. Four cases also have inconsistent physical queue inputs. The new transfer test reports numerical error directly; the historical prediction sheet remains in the supplement as a separate record.

Table 12
Retrospective transfer errors for BBRv3 condition means. MAE is the mean absolute error; normalized-goodput errors are fractions of capacity. Queue interpolation uses E1 only. The constant predictor uses the same calibration data. Each row evaluates four E3 conditions.

Evaluation group	Metric	Queue MAE	Constant MAE	Queue max. error
200 Mbps, actual $q = 1, 4$	u	0.00264	0.00264	0.00400
	J	0.01010	0.01134	0.01464
50 Mbps, actual $q = 4, 16$	u	0.00309	0.00309	0.00366
	J	0.02027	0.01444	0.02499

Table 13
Sensitivity of the retrospective E3 rate/fairness assessment to illustrative service requirements. Cells count runs meeting both limits, from 40 runs per algorithm. They are not independent condition counts.

Minimum u	Minimum J	BBRv3	CUBIC
0.90	0.95	40/40	27/40
0.95	0.95	0/40	27/40
0.90	0.99	38/40	19/40

8. Conclusion

This survey connects BBR version evolution, network conditions and performance objectives through a common assessment. Buffer depth, BDP and loss source organize the evidence; capacity, RTT, implementation, queue management and competing flows make each comparison interpretable. The analytical procedure translates fairness requirements into flow-share constraints and assesses utilization, delay, fairness and retransmission cost separately.

The 340-run evaluation makes the distinction between utilization and allocation concrete. In twelve BBR/CUBIC conditions, total goodput stays within 95.58–95.59 Mbps while the BBR share ranges from 0.143 to 0.327. Under unequal RTTs, measured long-RTT shares range from 0.769 to 0.969. The additional equal-RTT cases expose a throughput–fairness tradeoff: changing the stated service requirements changes which runs meet the joint assessment. These results place per-flow outcomes alongside the aggregate rate that otherwise conceals them.

For deployment assessment, the central lesson is to report the path and implementation conditions, measure each required outcome, and make service requirements explicit. The retained measurements and analysis support that assessment under the documented conditions. Learning-assisted control can be evaluated within the same structure by connecting its objectives and actions to measured allocation, delay and operating cost. A useful BBR comparison therefore explains both how much traffic the link carries and how that capacity is shared.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The revision includes new controlled-network measurements alongside published studies. Raw per-run configurations and logs, implementation manifests, analysis scripts and condition summaries are provided in

the accompanying experimental supplement. Search exports and screening records are provided in a separate search supplement. No public archive DOI is claimed.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of the original manuscript, the authors used OpenAI GPT Image 2.0 to generate Figures 3–6 and manually checked their technical content. During this revision, AI assistance was used for manuscript editing and preparation of plotting code. Figures 2–6 were redrawn using retained Python plotting scripts to correct delay labels, distinguish controller versions, and remove unsupported bandwidth-allocation inferences. The authors take responsibility for the manuscript content.

CRediT authorship contribution statement

Zonglin Feng: Investigation, Writing – original draft. **Ruohan Ma:** Investigation, Writing – original draft. **Shouzheng Xu:** Conceptualization, Supervision, Project administration, Writing – review & editing. **Xinyue Liu:** Investigation, Writing – original draft. **Tingjia Liu:** Investigation, Writing – original draft.

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